

ONE: Shaping the future of digital shipping with Google Cloud

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ONE leverages Google Cloud to improve business processes by deploying Google Workspace for employees to work anywhere and migrating SAP workloads to Google Cloud, and building a data lake on BigQuery to derive business insights from real-time data.

ABOUT OCEAN NETWORK EXPRESS



About Ocean Network Express

With its global headquarters in Singapore, Ocean Network Express (ONE) is a joint venture between "K" Line, MOL, and NYK. The shipping company serves more than 14,000 customers in 63 countries, supported by more than 7,000 employees. ONE provides container shipping services to 257 container ports all over the world. The company's fleet of 224 vessels includes some of the world's largest container ships, with a capacity greater than 24,000 TEU to transport large volumes of goods. ONE's vision

Google Cloud results

- Minimizes downtime by running mission-critical applications on Compute Engine in the cloud for 99.99% availability
- Simplifies archiving of email messages and files stored in Enterprise Vault for compliance needs
- Speeds up onboarding process with Forms so new hires can be productive from day one

Creating business collaboration with Google Cloud

is to be sustainable, resilient, and a trusted partner for customers in delivering global shipping solutions. The company won the EcoVadis Gold Medal award in 2019 for its commitment to sustainability.

Industries: Transportation

Location: Singapore

Apiqee API Platform

About partners

GoPomelo

Google Cloud Premier Partner GoPomelo delivers engineering and professional services to customers across Asia Pacific. With a proven track record in workplace transformation, GoPomelo helps companies improve productivity and collaboration with Google Workspace.

DXC Technology

DXC Technology helps global companies accelerate digital transformation by moving mission-critical workloads and applications to the cloud. The Google Cloud Managed Services Provider provides an extensive portfolio of services in more than 70 countries, which include infrastructure and application modernization, SAP on Google Cloud, smart analytics, contact center transformation, and DXC advisory and consulting services.

- Scales its ERP systems swiftly to meet customer demands and data volumes with SAP S/4HANA

At the port and on the high seas (<https://www.one-line.com/en>) star magenta containers and ships, a color against traditional blues and branding reflects the shipping company's creative and to differentiate it from competitors.

Formed in 2018 as a joint venture of Japanese shipping companies, including Kisen Kaisha ("K" Line), Mitsui O.S. Nippon Yusen Kaisha (NYK), ONE is a major container shipping company.

ONE embraces digital transformation with [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>) for its edge. The next-generation shipping enhances collaboration with [Google Workspace](https://workspace.google.com/) (<https://workspace.google.com/>), its running [SAP on Google Cloud](#) ([/sap](#)) unlocks data to drive deeper insights ([/bigquery](#)).

Deloitte

Deloitte is a leading global provider of audit and assurance, consulting, financial advisory, risk advisory, tax and legal, and technology consultancy. It was the primary partner on the Data Driven Enterprise for ONE.

Contact us (</contact>)

Google Cloud (<https://cloud.google.com/>)

SAP on Google Cloud
(<https://cloud.google.com/solutions/sap/>)

About partners

Apigee API Platform
(<https://cloud.google.com/apigee/>)

GoPomeio

BigQuery (<https://cloud.google.com/bigquery/>)

Compute Engine
(<https://cloud.google.com/compute/>)

With a proven track record in workplace

Cloud IAM (<https://cloud.google.com/iam/>)

transformation, our clients help

Cloud Storage (<https://cloud.google.com/storage/>)

It's time to get started with

Google Workspace
(<https://workspace.google.com/>)

DXC Technology

Gmail
(<https://workspace.google.com/products/gmail/>)

Google Drive
(<https://workspace.google.com/products/drive/>)

Cloud Managed Services Provider provides

Sites
(<https://workspace.google.com/products/sites/>)

than 70 countries, which include

Forms
(<https://workspace.google.com/products/forms/>)

smart analytics, contact center transformation, and DXC advisory and

"Ocean Network Express was for a company with a forward-thinking mindset. Google Workspace brings our different teams together to create new value for our customers. In real time, Google Cloud helps us achieve our goals with a scalable cloud platform for our various products, such as our digital quotation system, which speeds up responses to customers. Kentaro Yuasa, Deputy General Manager, Enterprise Architecture and Strategy, ONE.

Calendar

(<https://workspace.google.com/products/calendar/>)

Google Chat

(<https://workspace.google.com/products/chat/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Vault

(<https://workspace.google.com/products/vault/>)

Google
Workspace
changed our
mindset about
how we conduct
meetings.
Instead of
business travel,
we can easily
use Meet for
secure video
meetings to
discuss
challenges and
ideas as though

we're in the
same room."

—Lenny Lee, General
Manager, Groupware &
Collaboration, ONE Tech
Stop

Staying agile for business continuity during the pandemic

The COVID-19 pandemic amplified the importance of shipping to the smooth flow of raw materials, fuel, food, and medical supplies. From the get go, ONE helped minimize the impact of office closures and travel restrictions on its customers by having a business continuity plan in place using Google Workspace, in particular using [Gmail](https://workspace.google.com/products/gmail/) (<https://workspace.google.com/products/gmail/>) and [Google Meet](https://workspace.google.com/products/meet/) (<https://workspace.google.com/products/meet/>) to stay connected with customers, vendors, and colleagues, whether they were working remotely or in the office.

From prompt responses to sales inquiries from shippers, to the tight coordination of cargo between ports and countries, having secure access to Gmail and work data on Google Workspace means employees can deliver seamless customer service from any device.

ONE protects its employee's accounts and company data by enabling security features such as two-step

verification on Google Workspace. Furthermore, administrators use Cloud Identity and Access Management (/iam) to define access to work data on Google accounts even if employees log in with personal mobile devices. For employees who prefer to use a third-party email client, ONE set up Google Workspace Sync (<https://tools.google.com/dlpage/gssmo/>) to synchronize their Gmail and Google Calendar (<https://workspace.google.com/products/calendar/>) on the third-party interface.

"Google is flexible enough that we can allow people to use other tools while we continue to collaborate and innovate on Google Workspace," says Lenny Lee, General Manager at ONE.

Where face-to-face conversations are necessary, Meet becomes the default communication tool. It natively integrates with Google Workspace, so users don't need to switch between different applications to schedule meetings on Calendar, send invites with Gmail, and save recordings on Google Drive (<https://workspace.google.com/products/drive/>).

"Google Workspace changed our mindset about how we conduct meetings. Instead of business travel, we can easily use Meet for secure video meetings to discuss challenges and ideas as though we're in the same room," says Alan Sze, Deputy General Manager - BPIT at ONE.

ONE also uses Meet heavily for training purposes. GoPomelo (<https://www.gopomelo.com/>), the Google Cloud partner that set up Google Workspace for the

company, holds regular Meet sessions to help employees understand how to use Google Workspace to get work done in the most efficient way possible. Employees in different regions receive invites to fortnightly interactive video training and can view the recorded sessions in ONE's learning portal in [Google Sites](https://workspace.google.com/products/sites/) (<https://workspace.google.com/products/sites/>).

GoPomelo also helped the ONE team set up [Google Forms](https://workspace.google.com/products/forms/) (<https://workspace.google.com/products/forms/>) to be used for employee training. Now, they can easily collect information and gather feedback from employees during and after any training is conducted.



Setting employees up for success by automating the onboarding process

For a lean team of six IT staff in a rapidly growing company, having to manually onboard every new hire can be laborious and time consuming. With the help of GoPomelo and Google Workspace, the onboarding process is now automated. GoPomelo

wrote automated scripts to process access requests for software and backend systems such as its CRM (customer relationship management) for sales and its container solution for booking and documentation.

Using Forms, the HR representative only needs to fill up a new hire form to kickstart the onboarding process. This helps avoid unnecessary delays when the employee turns up on their first day of work. From IT setup to office equipment and system access, all the relevant departments are alerted and the appropriate documentation is complete.

Once onboarded, employees can manage their mailboxes efficiently on Gmail, so they can dive straight into their work instead of worrying about unnecessary administrative tasks like sorting out email folders. By default, email messages show by conversation threads so employees can read related messages in a single view. Instead of having to manually sort emails into folders and subfolders, users can assign a custom label to an email and create a filter to auto-tag similar messages with the same label. Employees can also use search with a combination of filters such as sender, attachment type, and date to quickly find any email or file they need on Drive.

Robust data archiving and discovery for compliance

Shipping is a highly regulated industry where shipowners operate by strict standards and government regulations. For compliance and

financial reporting, ONE needs to keep all of its business-related records for seven years. The company uses Vault (<https://workspace.google.com/products/vault/>), included with its Google Workspace Enterprise edition, to retain and search data.

The archiving process is transparent to users because emails are stored simultaneously in Gmail and Vault. This also makes it easy for the IT team to search and share employees' Gmail messages and Drive files when required for legal reviews. Even if employees delete emails and files from Google Workspace, all data remains in Vault.

"With unlimited storage on Google Workspace Enterprise, we don't have to worry about backing up email servers and restoring gigabytes of data from tape drives," says Lee. "It also helps us resolve any disputes quickly when they arise, as we are able to produce audit trails from Vault."

"One advantage of running SAP on Google Cloud is the business continuity on a highly available platform. Google

Cloud provides failover between sites so we can keep accounting tasks running even if a disaster occurs. We can also easily scale up and down to meet changes in business demands and data volumes. We can also carry out cloning of our environments easily if we need to perform system upgrades or enhancements"

—Alan Sze, Deputy
General Manager - BPIT,
ONE

Building high availability and disaster recovery for SAP in the cloud

As one of the largest container shipping companies globally, ONE handles large amounts of transactions with customers and vendors every day. The company uses SAP S/4HANA enterprise resource planning finance software at its global headquarters to manage revenues and payouts from its core operating system, including income from shipping contracts and truckers' payment for transporting containers.

Previously, ONE hosted an SAP accounting system locally in each country on its legacy cloud platform. The issue with that was the local accounting system could not easily integrate with the on-premises SAP S/4HANA at the global headquarters for a unified view of data for the whole company. To gain real-time financial visibility, ONE migrated all of its SAP workloads to [Compute Engine](#) (/compute). Today, the company leverages [API Management Platform](#) (/apigee) between SAP S/4HANA and uses third-party applications to determine exchange rates and make payments.

"One advantage of running SAP on Google Cloud is the business continuity on a highly available platform. Google Cloud provides failover between sites, so we can keep invoices and payments running even if a disaster occurs," says Sze. "We can also easily scale up and down to meet changes in customer demands and data volumes."

Google Cloud Managed Services Provider [DXC](https://www.dxc.technology/)
(<https://www.dxc.technology/>) guided ONE through the migration of SAP workloads to Google Cloud, from planning to testing and cutover, with zero downtime. The on-premises migration took three months, starting from the development environment, to the final production environment.

"Building a data lake is an example of how we're able to do more on Google Cloud compared to our existing on-prem infrastructure. We want to keep improving our platform capabilities with Google Cloud and deliver new solutions to grow our business."

—Kentaro Yuasa, Deputy
General Manager of
Enterprise Architecture
and Strategy & Innovation,
ONE

Gaining business insights with data-driven approach

Like most shipping companies, ONE's success is measured on its ability to improve profitability in the safest and most efficient way possible. As such, the company is accelerating the use of data analytics to enhance operational efficiencies and predict demand in the supply chain to quickly make business decisions. It plans to modernize its legacy data environment with a data lake on Cloud Storage with the help of Deloitte (<https://www2.deloitte.com/sg/en.html>), by the third quarter of 2021.

Unlike traditional data warehouses, a data lake can store both structured data such as tables and spreadsheets and unstructured data such as emails and pictures. This flexibility allows ONE to support future data models as the company evolves to meet new opportunities.

Currently, loading data from the SAP S/4HANA system to the legacy on-premises data warehouse takes a full day because it is unable to query tables that exceed 255 columns. The slow query performance means employees cannot make data-

backed decisions quickly. BigQuery, on the other hand, can query 10,000 columns per table to deliver business insights in minutes, which will be a game-changer when it comes to data-led decision-making. Apart from its SAP S/4HANA system, ONE will also synchronize data from its numerous applications with Google Cloud and transition its legacy data warehouse to BigQuery.

“Building a cloud-native data lake on Google Cloud follows our decision to use Google Cloud as the primary cloud where we’ve deployed key digital components such as Apigee API Management and Google Workspace,” says Tomohiro Nishikawa, Manager of BPIT Strategy and Innovation IT Modernization Strategy at ONE. “By centralizing the visualization and consumption of data in Google Cloud, we provide business users with fast access to trusted data to run their own queries.”

Moving forward, ONE wants to analyze data from its SAP S/4HANA and other data sources to uncover patterns and identify business value. For example, it can ensure more prompt payment for port charges so that its cargo can be released quicker from the ports, reducing unnecessary cost due to longer time spent at port. To optimize revenue, ONE can even analyze transaction patterns and boost sales with customers who are likely to pay on time.

“Building a data lake is an example of how we’re able to do more on Google Cloud compared to our existing on-prem infrastructure,” says Yuasa. “We want to keep improving our platform capabilities

with Google Cloud and deliver new solutions to grow our business.”